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RESEARCH-ARTICLE

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THINH DUC TRAN, Mittweida University of Applied Sciences, Mittweida, Sachsen, Germany

MANUEL HEINZIG, Mittweida University of Applied Sciences, Mittweida, Sachsen, Germany

HOLGER LANGNER, Mittweida University of Applied Sciences, Mittweida, Sachsen, Germany

MARC RITTER, Mittweida University of Applied Sciences, Mittweida, Sachsen, Germany

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Mittweida University of Applied Sciences

Published: 31 August 2025

[Citation in BibTeX format](#)

[MuC '25: Mensch und Computer 2025](#)
August 31 - September 3, 2025
Chemnitz, Germany

Towards Adaptive Difficulty and Personalized Player Experience: A Data-Driven Approach Combining Game Analytics, Sensor Data, and Personality Traits

Thinh Duc Tran

University of Applied Sciences Mittweida
Mittweida, Germany
ttran5@hs-mittweida.de

Holger Langner

University of Applied Sciences Mittweida
Mittweida, Germany
langner@hs-mittweida.de

Manuel Heinzig

University of Applied Sciences Mittweida
Mittweida, Germany
heinzig@hs-mittweida.de

Marc Ritter

University of Applied Sciences Mittweida
Mittweida, Germany
ritter@hs-mittweida.de

Abstract

Adaptive difficulty adjustment represents a critical mechanism for optimizing user experience in digital games. This study introduces a data-driven approach that integrates game telemetry, biometric sensor data, and personality profiles to classify player behavior, which can be used to inform adaptive artificial intelligence (AI) systems. A custom-built game prototype enables the collection of multidimensional data, including reaction times, accuracy, physiological responses (e.g. heart rate), and subjective stress assessments.

Unsupervised machine learning techniques, specifically K-Means and Agglomerative Clustering, were applied in multiple analytical iterations, first to segment players by performance and then by strategic behavior. The dataset was engineered to reflect gameplay efficiency and decision-making patterns, with preprocessing steps to minimize bias. Cluster validity was assessed using ANOVA, and feature relevance was evaluated through Random Forest analysis.

Results indicate that distinct clusters can be identified, each characterized by specific cognitive and strategic tendencies. These clusters display unique patterns across gameplay performance, physiological feedback, and subjective perception. The proposed classification model enables the foundation for personalized AI systems, capable of dynamic adaptation based not only on skill level but also on emotional and behavioral traits. The approach contributes to the intersection of game analytics, psychophysiological modeling, and adaptive gameplay design.

CCS Concepts

• **Computing methodologies** → **Machine learning algorithms**;
• **General and reference** → *Surveys and overviews*; • **Human-centered computing** → **HCI theory, concepts and models**;
Empirical studies in HCI.

Keywords

Adaptive Difficulty, Player Clustering, Game Analytics, Machine Learning, Player Behavior, Difficulty Adaptation, Video Game AI

ACM Reference Format:

Thinh Duc Tran, Manuel Heinzig, Holger Langner, and Marc Ritter. 2025. Towards Adaptive Difficulty and Personalized Player Experience: A Data-Driven Approach Combining Game Analytics, Sensor Data, and Personality Traits. In *Mensch und Computer 2025 (MuC '25), August 31–September 03, 2025, Chemnitz, Germany*. ACM, New York, NY, USA, 18 pages. <https://doi.org/10.1145/3743049.3743070>

1 Background and motivation

In recent years, the video game market has become an important part of the entertainment industry. As the industry has grown, the player base has also become more diverse, including everyone from casual players to professional e-sports athletes. This variety creates a challenge for game developers: they need to design games that appeal to many different types of players while still offering a balanced and enjoyable level of difficulty.

One key factor that influences player satisfaction is how well the game adjusts its difficulty. Many modern games still rely on fixed difficulty settings, which don't consider the individual abilities or preferences of each player. Dynamic difficulty systems, which adjust based on real-time analysis of player behavior, have the potential to make games more personal and engaging [26].

In addition to gameplay performance, personality traits and individual perceptions of difficulty also play an important role. Some players enjoy tough challenges and find them motivating, while others may feel frustrated if a game becomes too difficult. That's why it's important to consider not only objective gameplay data but also subjective factors like player preferences, stress levels, and playing style [10].

Due to advances in technology, it is now possible to use modern data analysis methods and external sensors to track and understand players' physiological responses. These tools open up new opportunities for adapting games dynamically, not just based on skill level, but also based on how players are feeling physically and emotionally [1, 17].

Even though static difficulty systems are still common in many games, they often fail to meet the needs of different player types.



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ACM ISBN 979-8-4007-1582-2/25/08

<https://doi.org/10.1145/3743049.3743070>

In contrast, dynamic difficulty adjustment (DDA) can make the experience more engaging by reacting to how a player performs. Table 1 shows the main differences between static and dynamic difficulty systems [26].

The goal of this research is to analyze player behavior to support the development of an AI-based system for dynamic difficulty adjustment. In modern video games, difficulty is not determined by a single factor but by the combination of many different in-game mechanics. Each game challenges players in different ways and across multiple skill areas, including both cognitive and motor skills. For example, real-time strategy games like *Starcraft* or *League of Legends* test players on two levels: mechanically—through actions like combo execution, aiming, and reaction time—and strategically—through resource management, game understanding, and decision-making [5].

A player's perception of difficulty often depends on their strengths and weaknesses in specific areas. Some players may struggle with fast reactions but excel at long-term planning, while others might be the opposite. In addition, personality traits also play an important role: some players enjoy high difficulty, while others prefer a more relaxed experience. Because of this, adaptive AI systems must first collect and analyze detailed data on player behavior to understand how different individuals play and what they need from a game [12, 26].

This study primarily investigates how player behavior can be categorized using telemetry and sensor data. The goal is to develop a framework for understanding player actions and patterns, which could potentially inform adaptive game systems. While the original scope also considered the dynamic adjustment of game difficulty and the role of physiological signals, the primary achievement lies in establishing a robust method for behavior classification.

Using a data-driven approach, players can be grouped by their playstyles, personality, and subjective feedback. These clusters make it easier to predict what difficulty level suits each player best and allow the game to adjust accordingly. By analyzing patterns in gameplay and feedback, adaptive AI can make smarter adjustments that create a more personalized experience. However, adapting difficulty on an individual level requires several technical steps and careful analysis.

As part of this research, a custom-built game prototype was developed to track and analyze player behavior. It combines telemetry data, like hit accuracy and reaction time, with sensor data, such as heart rate and self-reported difficulty. These combined data sources form the basis for future semi-automated AI systems that can make more accurate and effective difficulty adjustments.

The rest of the paper is structured as follows:

- Chapter 2 reviews related work on adaptive difficulty and game analytics.
- Chapter 3 explains the methodology, including data collection, clustering, and sensor integration.
- Chapter 4 presents the experimental findings.
- Chapter 5 discusses the results and outlines directions for future research.

Research Questions

Based on the outlined motivation and goals, this study is guided by the following research questions:

- (1) How can players be categorized using clustering techniques based on telemetry, physiological data and personality traits?
- (2) What differentiates the resulting player groups in terms of performance metrics, stress levels and strategic behavior?
- (3) How can these insights contribute to adaptive game mechanics that personalize difficulty based on individual player profiles?

2 Related Works

Adaptive difficulty and personalization of the player's experience have been widely explored in game research, particularly in the fields of game analytics, machine learning, and psychophysiological player modeling. Previous studies have focused on various aspects of player adaptation, including dynamic difficulty adjustment (DDA), clustering-based player classification, and the role of physiological and subjective data in difficulty optimization. This section provides an overview of existing research on adaptive game difficulty, data-driven player modeling, personality-based game adaptation, and the integration of physiological feedback in player analytics. By examining these works, we establish the foundation for our approach, which integrates telemetry data, biometric feedback, and personality traits to enhance adaptive difficulty systems.

2.1 Gaming Experience and Player Behavior

Understanding gaming experience and player behavior is essential for research in the area of digital games. Analyzing these aspects provides valuable insights into player interactions with the game [7]. Gaming experience encompasses the subjective experience of the player while playing. This experience is influenced by numerous factors such as emotional reactions, engagement, motivation, and flow states [4]. The concept of *Flow Theory*, introduced by Csikszentmihalyi, is of special importance. It describes the balance between challenge and player skill as a key determinant of engagement. During the flow state, players forget about the outside world and are fully involved in the game. Being in a flow state has massive positive affects such as intense focus on the activity, less distraction and the ability to use more cognitive energy to play the game and engage with the contents of the game.

2.2 Adaptive Difficulty Adjustment in Games

Dynamic Difficulty Adjustment (DDA) techniques aim to modify in-game challenges in real-time based on player performance [13]. Traditional rule-based approaches rely on pre-defined difficulty curves, while more advanced models employ machine learning and predictive analytics to provide a personalized player experience [23].

The cognitive load is a central factor in determining the quality of the gaming experience and how effective learning processes are accomplished in games. Adaptive AI can improve the learning process even further by adjusting the challenges to avoid cognitive overload. According to Sweller, it is of utmost importance, that the cognitive stays within an optimal range to ensure the most effective learning experience [22]. An adaptive AI could adjust the pace and

Table 1: Comparison of Static vs. Dynamic Difficulty Adjustment (DDA)

Feature	Static Difficulty Adjustment	Dynamic Difficulty Adjustment (DDA)
Adaptability	Fixed levels, does not change	Adjusts based on player performance
Player Engagement	Can be frustrating for players	Enhances player retention and engagement
Implementation Complexity	Simple, predefined difficulty	More complex, requires data analysis
Common Methods	Pre-set difficulty levels, manual selection	AI-driven adjustments, real-time telemetry analysis

complexity of games to fit the cognitive capacities of the player. This is intended to support learning experience, which has been associated with improved outcomes in prior studies.

One of the most well-known applications of DDA is the *AI Director in Left 4 Dead*, which dynamically adjusts the spawn rate of enemies and item availability based on player performance. Similarly, in *Resident Evil 4*, the game modifies enemy aggressiveness and ammunition availability depending on how well the player is progressing [6].

Despite their effectiveness, these methods often fail to consider player personality and subjective difficulty perception, which are critical in optimizing engagement and retention. Recent studies highlight the importance of incorporating telemetry data, physiological feedback, and player psychology into adaptive difficulty systems to enhance long-term player satisfaction [1, 2, 17, 19, 20, 24].

2.3 Measuring player behavior

In order to effectively analyze player data, relevant data needs to be gathered. A lot of data can already be extracted from the game itself in form of telemetry data. Game analytics is an established approach for understanding player behavior through telemetry data collection and pattern recognition techniques [25]. Through methods such as clustering and decision trees, players can be classified into various behavioral categories. But only internal game data is not enough to see the whole picture. In order to get more meaningful insights and more accurate correlations between game events and player behavior, external data needs to be collected through sensors like heart rate or oxygen saturation [19].

The heart rate can give a vague idea of how stressed a player is and can even be used to find the optimal difficulty and may support long-term player retention [19]. It is also important to keep a balance between challenge and well-being of the player. Excessive stress or heightened excitement can detrimentally influence performance and negatively impact the gaming experience. Research indicates that individuals in a state of relaxation typically attain superior outcomes [1].

Various methods are used to analyze the relationship between game events and time series data. The Pearson correlation coefficient is widely used, but is limited to linear correlations and normally distributed data [14]. The Kendall tau coefficient, which is suitable for non-normally distributed data, is more robust [15].

For more complex correlations, the nearest-neighbor-based method according to Luo et al. (2014) offers a more precise correlation between discrete events and time series [16]. In addition, recurrent neural networks (RNNs) enable the prediction of player behavior and the dynamic adaptation of game mechanics.

The choice of method depends on the application. While classical correlation analyses are suitable for simple patterns, modern AI approaches offer deeper insights and more flexibility for adaptive game systems.

2.4 Correlation between Personality and Player Behavior

When the goal is to personalize the gaming experience, one, often overlooked, key factor is the personality of the corresponding player. Different personalities have a multitude of preferences and learning styles. It is impossible to find a solution that fits all. In order to make the adaptive adjustments as accurate as possible, player personalities have to be taken into account [24].

The Myer-Briggs Type Indicator (MBTI) is commonly employed to examine player inclinations [3]. Extroverted individuals typically exhibit a preference for social engagement, whereas introverted individuals tend to favor solitary gaming experiences and intricate strategic elements [11]. Additionally, research suggests that certain personality types may exhibit superior performance in puzzle-oriented games compared to others. Furthermore, personality traits can serve as indicators of a player's preferred style of gameplay [8]. Such insights can facilitate the customization of the gaming experience to align with individual personality traits, thereby enhancing overall player enjoyment.

Although the potential of personality-driven adaptation is highlighted in the literature, the present study uses MBTI-based personality data only in an exploratory way due to limited sample size.

3 Methodology

In this study, a custom game prototype was developed to systematically capture player behavior and performance. The game was implemented using Unity Engine and designed to include both strategic and motoric challenges, allowing players to improve over time. To ensure the collection of high-quality and meaningful data, the game design incorporated elements that test decision-making, reaction speed, and execution precision.

Simultaneously, external physiological sensors were used to gather biometric data, providing deeper insights into players' emotional states and cognitive load. Specifically, heart rate- and oxygen saturation monitors were utilized to measure physiological responses during gameplay. To further enrich the dataset, subjective player experiences were recorded through standardized survey instruments. These surveys assessed perceived difficulty, stress levels, engagement, and strategic preferences, providing valuable

complementary data to the objective telemetry and biometric readings.

The collected data was analyzed using machine learning techniques, specifically K-Means Clustering and Agglomerative Clustering, to categorize players based on performance and strategic approach. By using ANOVA tests to examine the correlations between these clusters and their respective gameplay outcomes, a more precise and detailed representation of individual player behavior was established.

3.1 Development

Before developing the game, it was necessary to establish clear criteria to ensure alignment with the study's objectives. The primary goal of this research is to analyze player behavior based on multiple data sources, including telemetry data, physiological responses, and subjective survey feedback. These datasets serve as the foundation for an exploratory study aimed at identifying and categorizing distinct player behavior patterns. The focus lies in distinguishing various player types and understanding the underlying differences in their gameplay approaches.

3.1.1 Requirements for the game. To ensure the collection of rich and meaningful data, the game needed to provide interactive scenarios that are engaging while maintaining a low skill floor to accommodate a diverse player base. The inclusion of both casual and highly skilled gamers necessitated the development of a game environment that is intuitive, easy to learn, and quick to master. This design choice allows players across different skill levels to participate meaningfully, ensuring that gameplay differences are primarily driven by player tendencies and decision-making rather than by accessibility barriers.

While the game is designed to be accessible and easy to navigate, it must include strategic elements that provide players with multiple decision-making options. This approach aims to encourage diverse playstyles based on individual player choices and strategic preferences.

3.1.2 Game Description. The game presents a dynamic environment in which players control a character tasked with responding to incoming threats in the form of colored enemies, which appear at varying intervals. The core mechanic revolves around the use of two distinct weapons, each corresponding to a specific enemy color. Players must rapidly identify the appropriate weapon for an approaching enemy and execute the correct action in response.

The game mechanics are intentionally designed to be intuitive yet strategically engaging, ensuring accessibility while maintaining a cognitively demanding challenge. Players utilize a red weapon on the left, fired using the left mouse button, and a blue weapon on the right, activated with the right mouse button. Colored enemies appear periodically, and the player's objective is to eliminate as many as possible to accumulate points. The point system rewards accuracy and efficiency, with smaller enemies yielding higher point values upon successful hits.

Beyond the standard red and blue enemies, additional enemy types introduce greater complexity and strategic depth. Purple enemies require a shot from both weapons to be eliminated but provide increased point rewards. Additionally, yellow enemies feature a

countdown timer displayed above them, and if the timer reaches zero before the enemy is destroyed, the player loses one life. The game ends when all available lives are lost, reinforcing the necessity for quick reflexes, precise decision-making, and adaptive strategy.

To introduce an additional layer of resource management, each weapon has a limited ammunition capacity of 10 shots, which automatically refills every second. However, if a player's ammunition is completely empty, the weapon enters a three-second reload phase as punishment. This mechanic encourages players to balance rapid shooting with ammunition management, particularly in high-intensity situations.

Strategic gameplay is further enhanced by power-ups in the form of chest items scattered throughout the game. When these chests are destroyed, they reveal a star power-up, which provides a reload speed boost to the weapon that was used to shoot it. This mechanic introduces risk-reward dynamics, as players must decide whether to momentarily shift focus from enemies to obtaining power-ups that could offer long-term advantages.

Each game round has a maximum duration of 180 seconds, with increasing difficulty as the round progresses. Forty seconds before the round ends, a boss enemy appears, who has a large health bar. Players deal minimal damage to the boss through normal attacks, but there is a specific mechanic designed to defeat it. During regular gameplay, shooting red and blue enemies gradually charges up a special weapon bar. Once this bar is fully charged, the corresponding weapons become significantly more powerful, allowing the player to inflict substantial damage on the boss. Successfully defeating the boss rewards the player with a substantial point bonus, providing a motivation for players to strategically charge their weapons while managing regular enemy threats. A Screenshot of in-game footage can be seen in Figure 1. The game features colored enemies and chests, with the player managing two weapons and their respective ammunition. Player lives are displayed in the top left, while score, accuracy, and remaining time are shown on the right.

By integrating these diverse challenges, resource management mechanics, and dynamic difficulty elements, the game serves as an ideal testing environment for studying player behavior, strategic adaptation, and decision-making under pressure. The combination of time-limited gameplay, weapon management, power-ups, and end-game boss mechanics ensures that different playstyles can develop, making the collected telemetry data highly valuable for further analysis and classification.

3.2 Data Collection

Figure 2 provides an overview of the data collection pipeline used in this study. The data collection was structured around three primary sources: telemetry data, physiological data, and subjective survey responses. These data streams were integrated to provide a multidimensional perspective on player interaction, physiological states, and subjective perceptions. This holistic approach ensured a comprehensive understanding of player behavior.

3.2.1 Telemetry Data. Telemetry data serves as the foundation for understanding player behavior in a quantitative manner. It includes detailed in-game event logging, which captures every action a player takes during gameplay. Specifically, the game was

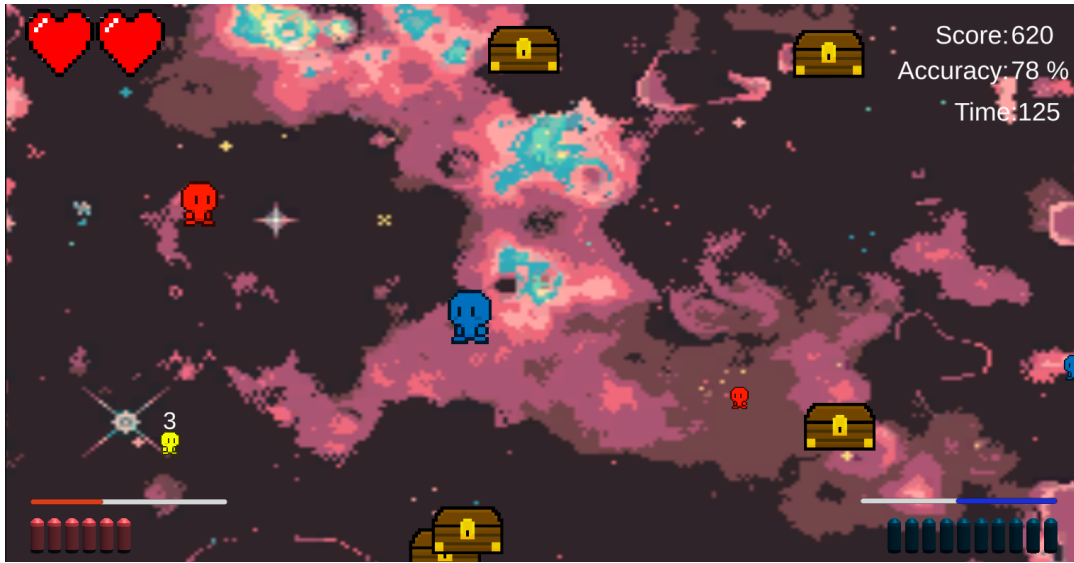


Figure 1: Gameplay Interface

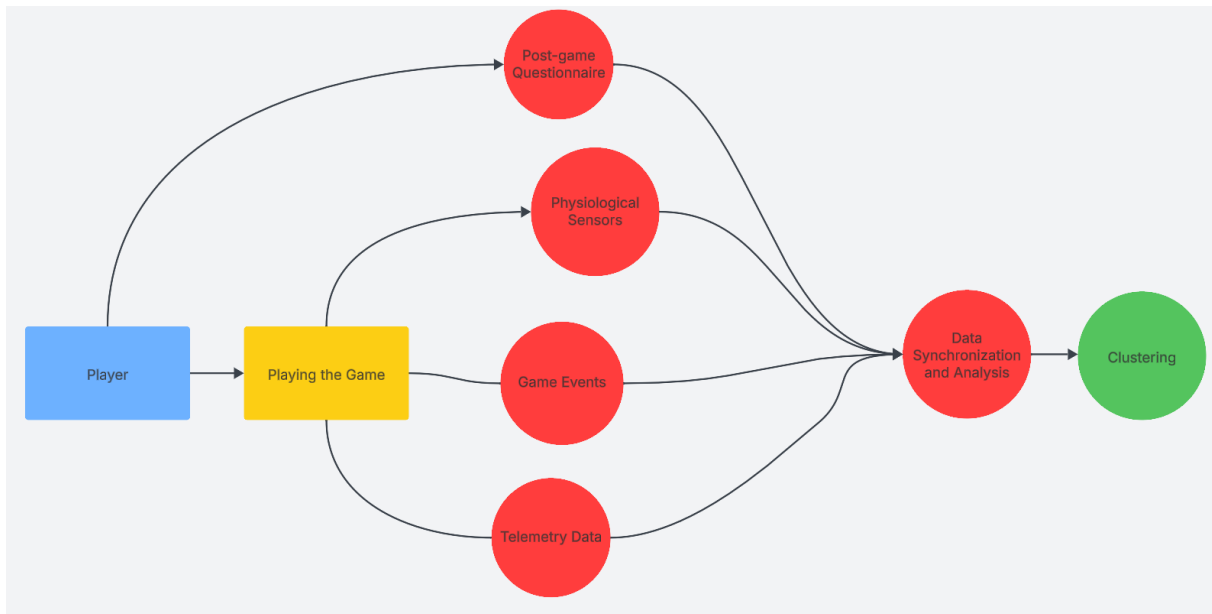


Figure 2: Overview of the data collection pipeline used in the study.

designed to record key performance indicators such as reaction times, hit accuracy, movement patterns, and decision-making speed. These metrics were collected in real-time and stored with precise timestamps to ensure accurate synchronization with physiological data.

To achieve meaningful analysis results, the telemetry data was structured into event-based logs that mapped player inputs to their corresponding in-game actions. This allowed for tracking strategic decision-making and motoric execution, two core components of player performance. For example, a player’s reaction to an in-game

challenge, such as an enemy encounter, could be analyzed by measuring the time between stimulus presentation and the player’s response. These data points provided an objective basis for clustering players into distinct categories based on their behavioral tendencies.

3.2.2 Physiological Data. In addition to telemetry data, physiological responses were monitored using external biometric sensors to gain deeper insights into players’ cognitive and emotional states.

Specifically, heart rate and oxygen saturation levels were continuously recorded throughout gameplay. Especially the heartrate is a well-established indicator of stress, engagement, and cognitive load, making them valuable for understanding how players react to different challenges within the game [1, 22].

Although the connection between Heart rate variability (HRV) and cognitive abilities is not well understood, studies show that HRV could serve as a promising factor when evaluating cognitive performance. Rapid fluctuations in heart rate can indicate a high level focus or stress, while more stable patterns may suggest a relaxed or disengaged state [9]. By correlating these biometric signals with in-game events, it was possible to determine how different gameplay situations affected individual players on a physiological level.

To ensure accurate synchronization between physiological data and in-game events, a real-time timestamping system for in-game events was implemented. The monitor hardware logged every physiological data point relative to a start time. This can then be used to sync with a timeline of game events. For instance, when a player encountered a high-stakes decision or an intense action sequence, the corresponding physiological response was captured and linked to that exact moment in gameplay. This approach enabled a precise analysis of how players' physiological states fluctuated in response to varying gameplay conditions.

3.2.3 Subjective Survey Responses. While telemetry and physiological data provide objective insights, they do not fully capture the player's subjective experience and perceived difficulty. To address this, standardized post-game surveys were administered to gather self-reported player feedback. These surveys were designed to assess multiple aspects of the gaming experience, including perceived difficulty, stress levels, engagement, and strategic preferences.

The survey questions were structured using Likert scales and open-ended responses to ensure both quantitative and qualitative insights. Players were asked to rate the perceived challenge of different in-game scenarios, their level of enjoyment, and their strategic approach to overcoming obstacles. Additionally, they provided information about their prior gaming experience and familiarity with similar game mechanics. This self-reported data was essential for contextualizing the telemetry and physiological findings, as it allowed for an examination of how objective performance aligned with subjective perceptions.

3.3 Evaluation Process

The evaluation of the study followed a structured process to make sure the collected data was reliable and consistent. Participants were recruited through social media and personal invitations, starting about three weeks before testing began. One week before their session, participants who hadn't already completed the Myers-Briggs Type Indicator (MBTI) test were asked to do so. This helped include personality data in the analysis without using up time during the main experiment.

Before playing, participants were fitted with biometric sensors: a heart rate monitor worn on the arm and an oxygen saturation sensor worn as a ring. These sensors were designed to be comfortable not interfering with gameplay. A short calibration period was used to make sure the sensors recorded data accurately. Participants

also received instructions on the controls and goals of the game to reduce confusion and ensure everyone started with the same understanding.

During the gameplay phase, players completed multiple rounds using the custom-built game prototype. While they played, their performance, decisions, and physiological responses were recorded. Telemetry data such as reaction time, accuracy, and player behavior was collected throughout the session. In addition to the in-game data, players were also asked to report their own experiences, including how stressed or engaged they felt, and how much they enjoyed the game.

After playing, participants took part in a structured interview to share their personal impressions. The questionnaire asked about their experience with the game, their stress levels, and their strategies. This feedback helped to add context to the data collected during gameplay and made it easier to understand the players' behaviors.

To keep the results consistent, the tests were run using the same hardware setup for everyone. Afterwards, the data was cleaned and prepared for analysis. Key variables like accuracy, survival time, reaction time, and decision-making patterns were examined and used in the clustering process. This thorough evaluation helped build a clear picture of how different types of players approached the game and what factors influenced their performance and experience.

3.4 Player Classification and Clustering

To identify distinct player behaviors and performance patterns, machine learning techniques were employed to classify and cluster players based on their in-game actions, physiological responses, and subjective feedback. The goal of this classification was to create a more accurate and detailed representation of individual playstyles by integrating multiple data sources. By analyzing correlations between these datasets, it became possible to distinguish meaningful differences in how players engage with the game, make decisions, and perform in various game-related tasks.

The classification process began with preprocessing the collected data to ensure consistency and comparability across players. Telemetry data, such as reaction times, hit accuracy, and chosen targets were transformed in a tabular dataframe for further analysis. Physiological data, including heart rate variability and oxygen saturation levels, was aligned with in-game events to capture real-time emotional and cognitive responses. Subjective survey responses were transformed into numerical scales to facilitate integration with objective data. A cutout of the tabular data can be seen in Table 2.

Once the data was preprocessed, clustering algorithms were applied in two iterations to identify distinct player groups. The first iteration focused on classifying players based on their performance. For an adaptive AI system to make precise adjustments to difficulty, it is essential to accurately determine the player's skill level. A study by Silva et al. (2015) explored using performance metrics, such as player success rates and in-game behavior patterns, to dynamically adjust game difficulty, highlighting the importance of precise skill level assessment [21]. The objective was to categorize players into different groups based on their overall success

PlayerID	Level	RoundID	RoundNumeric	Score	RoundTime	Hitrate
IACH	Level3	12_34_23	45263	1010	76.66615	0.819149
IACH	Level3	12_35_58	45358	1050	87.00468	0.828571
IACH	Level3	12_37_34	45454	495	60.44731	0.648336
IACH	Level3	12_38_43	45523	1590	111.27910	0.815385
IACH	Level3	12_40_47	45647	980	68.75993	0.824176

Table 2: Excerpt from the gameplay dataset. Each row represents one gameplay round. The PlayerID was freely chosen by the participants. The remaining columns contain performance metrics such as score, round duration, and hit rate, as well as information about the game level and unique round identifiers.

while also considering the specific metrics that contributed to their success.

The second clustering iteration aimed to identify different strategic approaches and behavioral patterns, independent of overall success. Rather than focusing on performance outcomes, this step analyzed how players interacted with the game and the patterns that emerged in their decision-making processes. To ensure fairness in the classification process, biases in the dataset were adjusted. For instance, instead of using the absolute number of successful hits, the hit accuracy percentage relative to total shots fired was used to normalize differences between players who survived longer and, therefore, had more shooting opportunities. This adjustment ensured that clustering results were not disproportionately influenced by survival duration but instead reflected genuine behavioral tendencies.

To conduct the clustering analysis, K-Means was used as the main method because it is efficient for grouping players based on numerical performance data. The algorithm grouped players who showed similar patterns in reaction speed, strategy, and physiological responses. Since K-Means requires the number of clusters to be set in advance, Agglomerative Clustering was also used. This method helped to explore the relationships between players and find the number of clusters that best matched the structure of the data. Using both methods made the classification more reliable and gave useful insights into player preferences, playstyles, and decision-making strategies—information that could help shape future adaptive game systems.

After the clustering, statistical tests were run to check if the differences between the groups were meaningful. ANOVA tests were used to compare performance metrics, behavior patterns, physiological responses, and how difficult the players said the game felt. This helped make sure that each group reflected a real type of player behavior and wasn't just random variation in the data. The results showed that players could be grouped based on their strategies, skill levels, and how engagement during the game.

These findings suggest that combining different types of data together may improve the accuracy of player classification compared to only looking at performance scores. Some players had similar performance levels but were still grouped differently because of how they reacted physically or how they experienced the game. This highlights the importance of also looking at emotional and cognitive factors. Overall, the results offer a strong foundation for future research on adaptive game design, where difficulty and game mechanics could be adjusted based on a deeper understanding of how players behave and feel while playing.

4 Results & Analysis

The experimental sessions were conducted over a one-week period, with each participant engaging in the study individually. Each session lasted approximately 45 minutes and consisted of three key phases: preparation, gameplay, and post-game assessment. A total of 13 participants took part in the study, comprising four female and nine male players, aged between 20 and 26 years, with a mean age of 23 and a standard deviation of 1.68.

The dataset consists of telemetry data, physiological measurements, and subjective player feedback. Telemetry data includes in-game actions such as accuracy, reaction time, survival duration, and weapon usage. Physiological data, collected through external sensors, provides insights into player engagement and stress levels. Subjective feedback, gathered through surveys, captures perceived difficulty, strategy preferences, and overall experience.

4.1 Player Performance and Strategy

The analysis of player performance and strategy aimed to identify distinct player groups based on their success and behavioral tendencies. Two primary clustering iterations were conducted: the first focused on quantifying overall player performance, while the second examined strategic decision-making and behavioral patterns. Following these clustering processes, statistical tests were performed to validate the findings and assess the significance of the differences among the identified groups.

Each clustering iteration was conducted in multiple steps. First, the elbow method was applied to determine the optimal number of clusters. Subsequently, K-Means clustering was used to segment players into distinct groups based on their performance or strategic behavior. To complement this, Agglomerative Clustering was employed as a hierarchical, bottom-up approach that does not require a predefined number of clusters. The resulting dendrogram was analyzed to extract the optimal number of clusters, allowing for a comparative evaluation between the two clustering methods. This comparison facilitated an assessment of consistency in player segmentation and helped identify the method that yielded the most meaningful results.

Before analyzing, it is important to select only the relevant data. The quality of the data decides how good the results will be. Therefore, the dataset has to be adjusted to the goal of the clustering.

4.1.1 Clustering Based on Performance. The first clustering iteration aimed to group players based on their overall success in the game. Instead of relying only on total scores, the clustering algorithm included several performance metrics, such as average

survival time, hit accuracy, reaction time, and how efficiently players managed their resources. Some players achieved high scores through accurate shooting, while others focused more on staying alive and using their resources carefully. By combining different aspects of success, this clustering approach gave a broader view of player skill and the different ways players approached the game. It is notable, that most of the data in this dataset are absolute values like total hits or survival time. This allows for a direct comparison between players performance.

First, the elbow method was used to determine the optimal number of clusters. For that, multiple iterations of K-Means Clustering were performed and the sum of the inertia was plotted. The elbow graph with an elbow at a clustercount of three can be seen in Figure 3.

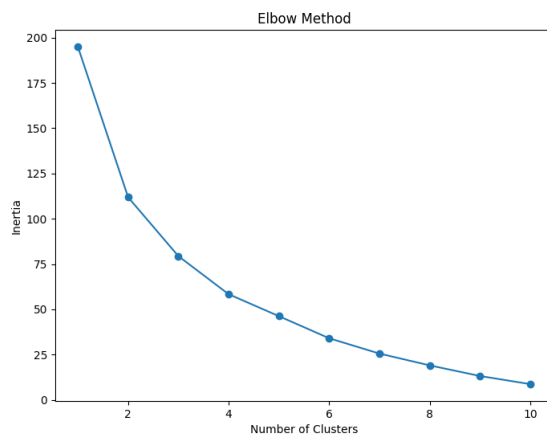


Figure 3: Course of inertia with increasing number of clusters. An “elbow” can be observed at a cluster count of three.

To better understand which factors influenced these clusters, a Random Forest classifier was used to analyze feature importance which can be seen in Figure 4. As shown in Figure 4, *Roundtime* (how long a player survived) and *score* are among the top 10 features with the highest importance. These two metrics were key to distinguishing between performance groups.

After clustering, the resulting clusters of players were compared. The survey data and cluster distributions did not show strong direct correlations, but there were some interesting trends. Cluster 1 mostly included experienced players who reported higher enjoyment and tended to prefer more relaxed gameplay. Cluster 0 included players who either had to exert more effort or actively sought out more intense challenges. Cluster 2 consisted mainly of less experienced players with moderate enjoyment, preferring a balanced level of difficulty.

Figure 5 shows a comparison of the top 10 performance features. Cluster 1 includes the best-performing players, Cluster 0 represents average performers, and Cluster 2 contains those with the lowest scores. Interestingly, Cluster 2 had longer survival times than Cluster 0 but still earned fewer points. This suggests different gameplay strategies: players in Cluster 2 focused on minimizing

risk and maximizing survival, while players in Cluster 0 were more efficient at point collection but took higher risks.

Power-up usage shows a similar trend. Although Cluster 2 players survived longer and encountered more power-ups, they collected fewer than those in Cluster 0, indicating a lower perceived value or level of engagement with the game mechanics.

Figure 6 provides a compact overview of the normalized mean values for each cluster. Cluster 0 players had short round times and minimal heart rate changes, possibly indicating lower engagement. Cluster 1 players performed the best, used power-ups frequently, and showed the highest heart rate changes—suggesting intense focus and gameplay engagement. Cluster 2 players had low scores, moderate survival times, and lower heart rate changes, pointing to a more passive or cautious playing style.

Furthermore, Agglomerative Clustering was used as an additional hierarchical clustering method. Unlike K-Means, this technique does not require the number of clusters to be known in advance. Instead, the optimal number of clusters can be identified by analyzing the resulting dendrogram, which was created using the *ward* method and is shown in Figure 7 [18]. The dendrogram shows a clear separation between two main cluster groups (orange and green), marked by the blue branches. These main clusters also contain subclusters that highlight further differences within each group. This type of visualization helps to better understand how similar or different players are, and allows deeper analysis depending on how detailed the grouping needs to be.

As expected, *Score* and *RoundTime* were among the most influential features, confirming that the clustering based on performance worked as intended. The boxplots in Figure 8 further show clear differences between strong and weak players.

Cluster 0 players tended to have lower scores, shorter round times, and less frequent power-up usage. Their heart rate changes were minimal or even below their resting levels, possibly indicating low stress or engagement. Cluster 1 players had higher scores, longer round times, and consistently used power-ups. Their heart rate changes were also more pronounced, suggesting higher intensity and focus.

In summary, both K-Means and Agglomerative Clustering effectively distinguished players based on their performance. These clustering methods could be useful in future adaptive systems to classify new players early and tailor the difficulty accordingly.

An approach based on clustering can help game developers identify different skill profiles within their game. This makes it possible to design targeted tools that support players in improving specific abilities. For example, a player who survives for a long time but scores low might benefit from learning more proactive strategies to increase their points. On the other hand, players with high scores might need better resource management to maintain their performance throughout the game. While these examples are based on a simplified prototype, a complete game would involve a wider and more complex range of skills. Grouping players by their strengths and weaknesses could lead to a more accurate and personalized way of supporting skill development.

4.1.2 Clustering Based on Strategy. The second clustering iteration aimed to group players based on their strategic behavior, rather than just performance outcomes. To make sure the results weren’t

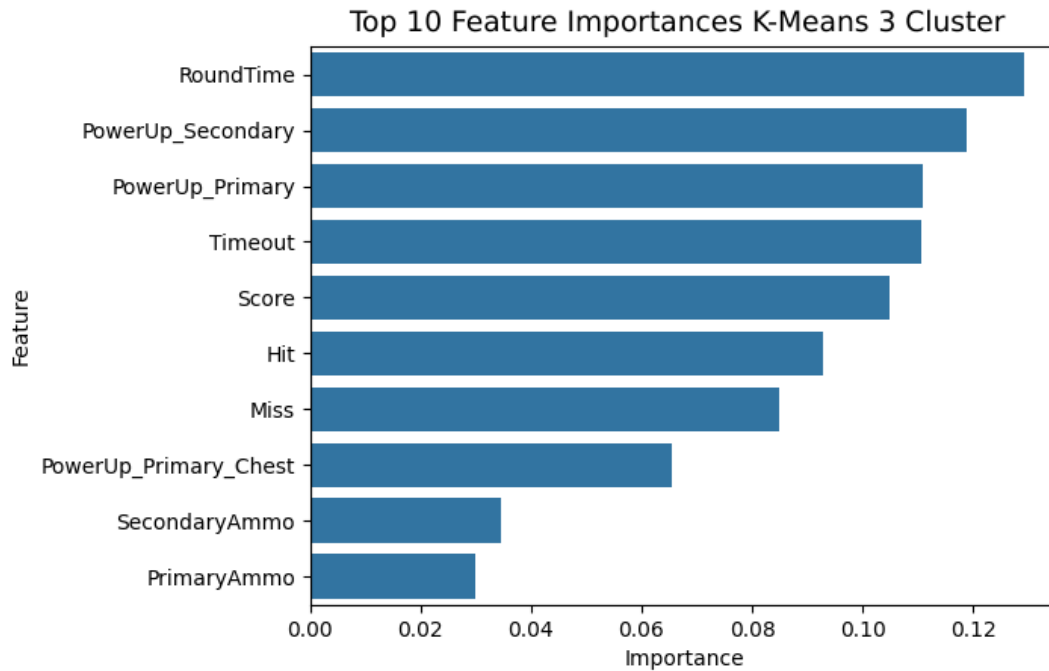


Figure 4: The top 10 most important features when clustering based on performance

biased by differences in playtime or scoring, absolute values were converted into relative ones. For example, instead of using the total number of hits, the hit rate (successful hits compared to total shots) was used. This was important because players who survived longer naturally had more actions and events, but that doesn't necessarily mean they had a better or different strategy. So the clustering focused on proportions and tendencies rather than raw numbers. This normalization helped ensure that all players were treated fairly in the analysis, no matter how long their rounds lasted.

This clustering approach is different from most related studies, which usually focus on how well someone plays (like score or win rates). Instead, this method looks at how players choose to play, which offers a new perspective on playstyles and preferences. It helps uncover behavioral patterns that aren't visible when only looking at performance data.

Players were grouped into strategic clusters, mainly separating aggressive and defensive styles. Aggressive players tended to use up their ammo quickly to eliminate as many enemies as possible. Defensive players, on the other hand, tried to survive longer and use fewer resources. This helped reveal different ways of thinking and reacting to game situations.

The feature importance analysis showed that there are four features that had the major influence on the strategy-based clusters. All of these features were related to ammunition management, such as the use of power-ups for both weapons and the average ammo usage for primary and secondary weapons. These features had the highest importance scores (around 0.11). After ammo usage, the portion of combo monsters eliminated had the next biggest influence,

which suggests that some players focused more on these special enemies than others.

Figures 9 and 10 show some key differences between the clusters. One important observation is that Clusters 0 and 3 each contain only one player, as indicated by the single line in the boxplot rather than a box. This means that the results from these clusters may not be as generalizable due to the small sample size.

Cluster 2 players mainly used their primary weapon and barely used the secondary one. Because of that, they collected fewer secondary power-ups and focused more on primary ones. This shows a clear single-weapon strategy, which most likely was chosen to reduce the cognitive load by removing the need to switch between weapons constantly and therefore being able to focus on aiming. The other clusters were more balanced in their weapon use, with a more even distribution of power-ups.

Combo monster targeting was also different across the groups. Players in Clusters 0 and 2 didn't focus much on combo monsters, while those in Clusters 1 and especially 3 gave them more attention for extra points. This lines up with earlier results from performance-based clustering, where some players played more for points, and others focused more on survival.

The heatmap also shows that Cluster 0 had a high focus on blue monsters, while Cluster 2 mostly ignored them. This fits with the earlier observation that Cluster 2 didn't use the secondary weapon much. Cluster 2 also reported the highest level of subjective stress, which might be linked to their more focused playstyle.

Clusters 1 and 3 included several players and are easier to compare. Cluster 3 had a higher focus on combo monsters and slower reaction times, which could suggest a more careful or strategic approach. Cluster 1 had faster reactions and used fewer shots to open

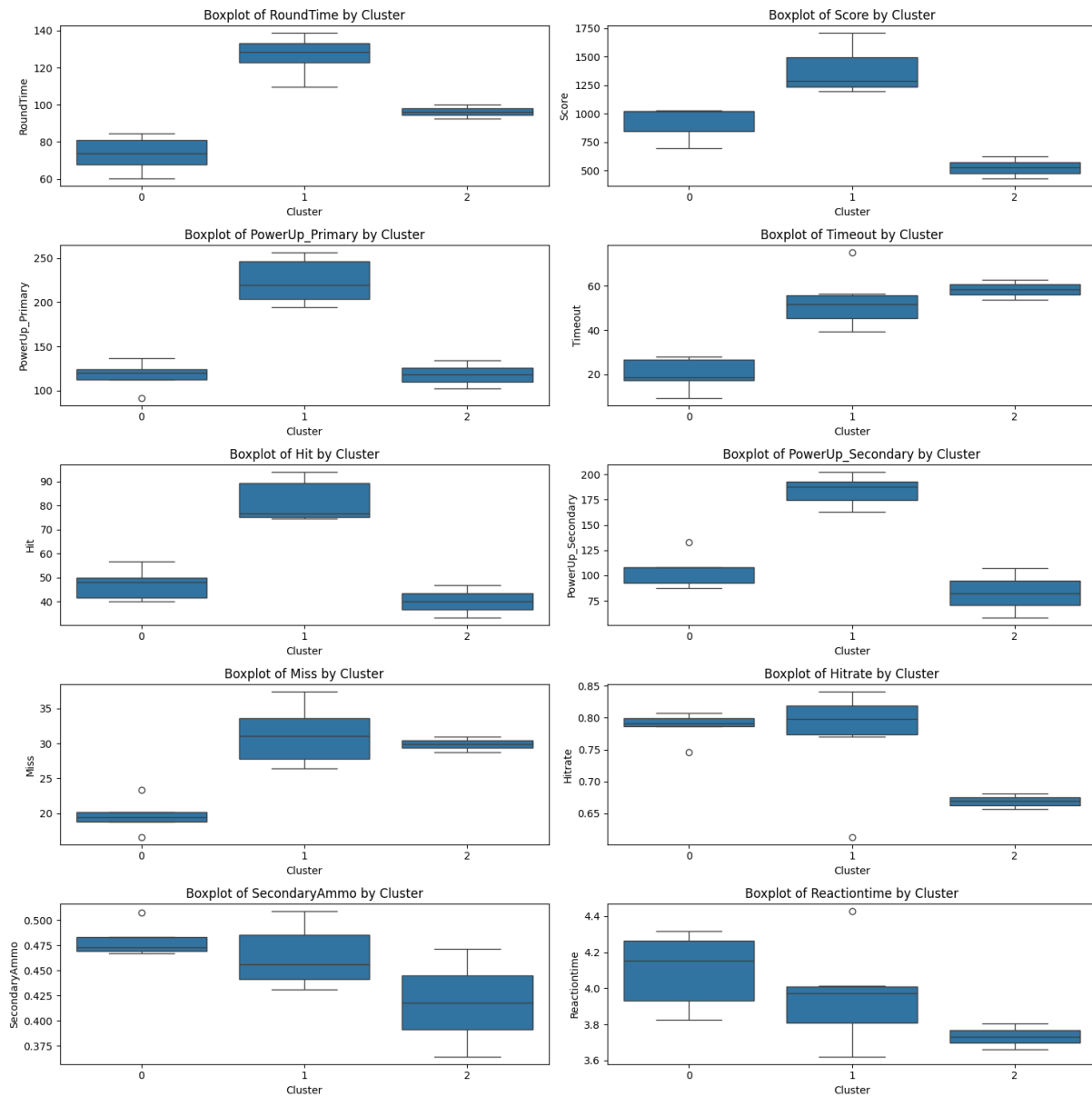


Figure 5: Boxplot comparison of the three performance-based player clusters. Cluster 1 represents high performers, Cluster 0 average performers, and Cluster 2 low performers, with noticeable differences in survival time and score efficiency.

chests with the primary weapon, probably to save ammo. They also used both weapons more flexibly, while Cluster 3 mainly used one weapon for specific tasks like defeating bomb monsters. This shows different personal preferences or habits when choosing weapons. Players in Cluster 3 also reported more changes in their stress level, while those in Cluster 1 had more stable and lower stress, which could mean they were more stress-resistant.

To look deeper into the player differences, a dendrogram was created using the Ward method (Figure 11). It shows how similar the players are in their behavior. While one player (RASU) stood

out clearly, a better split was found at a slightly lower level in the diagram, resulting in four meaningful clusters.

Player cluster assignments:

- **Cluster 0:** {RASU}
- **Cluster 1:** {RADI, ZAMI, IACH, IEMA, IACH_D, RAWI, RRBO}
- **Cluster 2:** {ZNHA}
- **Cluster 3:** {LNDR, RPNÜ, RLFR, TLCO}

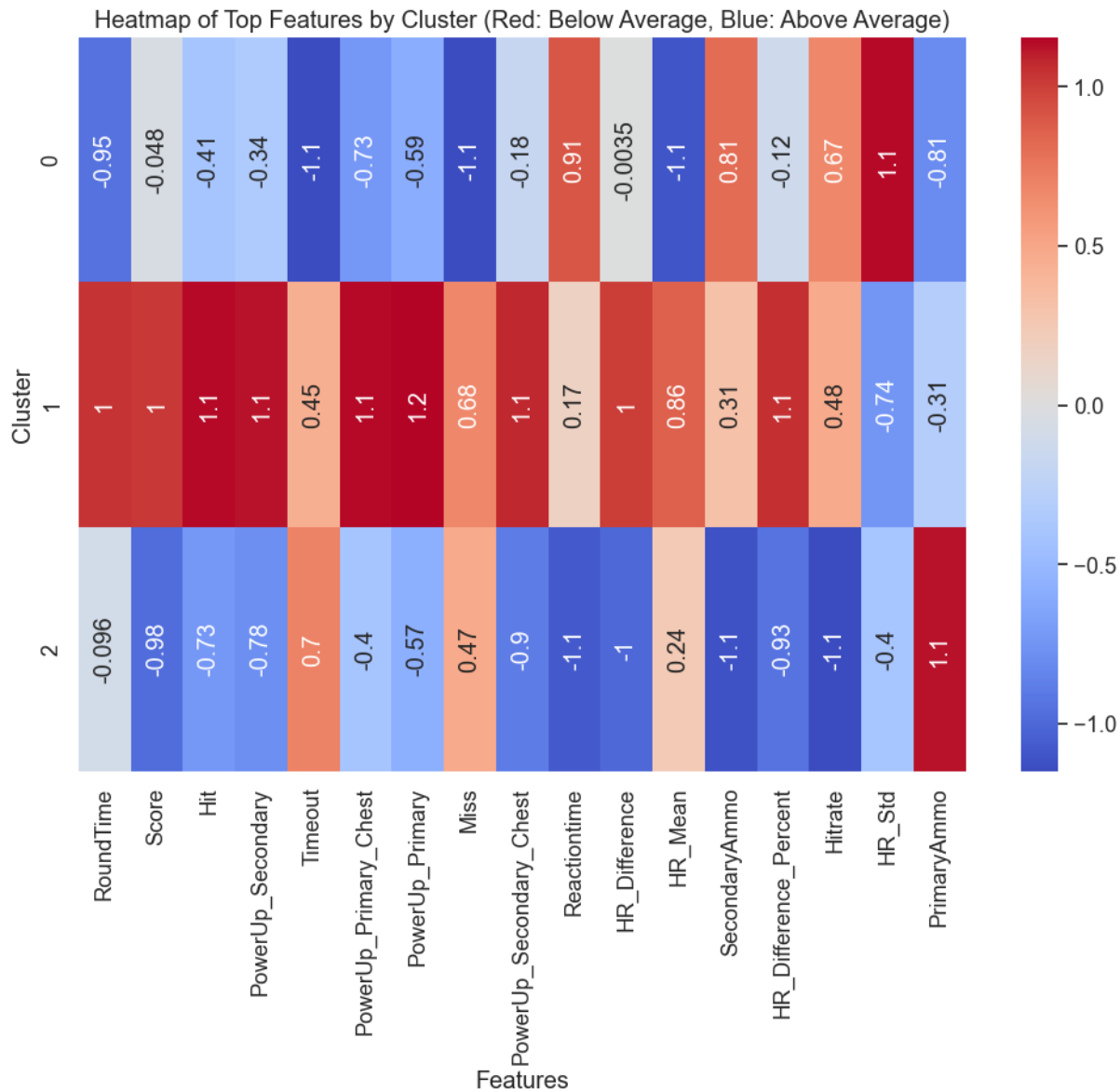


Figure 6: Heatmap comparing normalized mean values of key performance and physiological features across the three player clusters. Blue indicates above-average values, red below-average.

The results from the dendrogram matched the K-Means clusters, which means that the players were grouped in a stable and consistent way based on their gameplay data.

Since the results from Agglomerative Clustering were almost the same as those from K-Means—including the feature importance and cluster differences—no further analysis of those results was included here.

Findings like this can help game developers better understand what kinds of strategies players enjoy and what mechanics are being used or avoided. Instead of only adjusting difficulty, games could also adapt to personal preferences. For example, action-oriented

players could get more enemies, while those who like exploration could find extra hidden rewards.

This strategy-based clustering adds something new to existing research. It suggests that analyzing strategic choices may provide additional insights into player preferences beyond performance alone. This opens the door to new ways of making games more personal and engaging.

4.1.3 Correlation between Player Types. To assess the relationship between different player types and their performance, statistical tests were conducted to determine significant differences between groups. The clustering results, which segmented players based

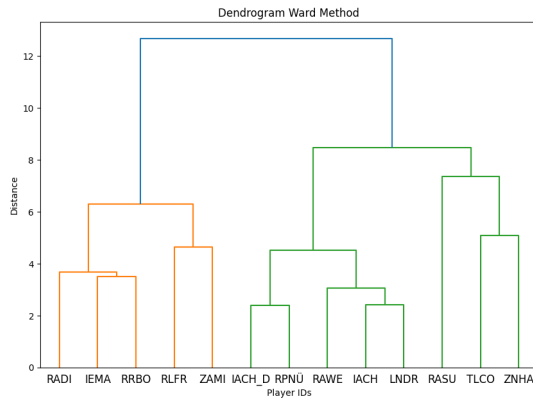


Figure 7: Dendrogram with Ward Method for hierarchical clustering. The diagram visualizes player similarities based on performance metrics. Lower branches indicate higher similarity. The further up the branches go, the more difference between clusters there are.

on performance and strategy, served as the foundation for this analysis. The goal was to identify whether specific playstyles were statistically linked to higher or lower performance outcomes or if highly successful players followed similar strategies.

The ANOVA test was employed to compare the means of multiple groups and examine whether distinct player clusters exhibited significant differences in gameplay metrics such as score, survival time, accuracy, and weapon usage. Before conducting the ANOVA test, a Levene's test was applied to verify the assumption of homogeneity of variances across the clusters. If the Levene's test confirmed equal variances ($p \geq 0.05$), a one-way ANOVA was performed. In cases where the ANOVA test returned a significant result ($p < 0.05$), a post-hoc Tukey-HSD test was conducted to identify specific differences between groups.

In case the Levene's test indicated unequal variances ($p < 0.05$), the non-parametric Kruskal-Wallis test was used instead. This test is better suited for datasets where the assumption of normality is violated. If the Kruskal-Wallis test identified significant differences, a Dunn's post-hoc test was performed to pinpoint which groups differed from one another.

The results of the statistical analysis revealed several significant differences between player clusters. As shown in Figure 12, the heatmap visualizes the results of the ANOVA and Kruskal-Wallis tests across various gameplay metrics. Darker colors represent more significant results ($p < 0.05$), indicating that the differences observed between clusters are statistically meaningful.

Statistical Highlights:

- **Reaction Time:** The figure indicates that reaction time significantly differed between player clusters ($p = 0.036$, ANOVA). Lower-performing players exhibited slower reaction times compared to high-performing groups, suggesting that faster reaction speed is an important factor for success in the game.

- **Enemy Elimination:** The heatmap shows a highly significant difference in the proportion of combo monsters eliminated between clusters ($p = 1.9e-13$, ANOVA). Higher-performing players consistently targeted these high-value enemies, reflecting an aggressive strategy to maximize their score. In contrast, differences related to red monsters (RedMonster_Normal and RedMonster_BullsEye) were not statistically significant ($p = 0.073$), indicating that not all enemy types were targeted equally by successful players.
- **Physiological Differences - Stress Levels:** Significant physiological differences were also observed. Both *StressLevel* ($p = 0.00014$) and *StressVariance* ($p = 0.00017$) showed notable differences, confirming that players in higher-performing groups experienced higher stress levels and more stress variability. This suggests that stronger players played more intensely and were more reactive to stressful situations. Higher stress levels may also be associated with greater engagement and emotional investment in the game.
- **Shooting Frequency:** The analysis revealed that players who performed well fired significantly more shots per ten seconds ($p = 5.6e-06$). This pattern is consistent with an aggressive playstyle and may reflect an attempt to minimize downtime from reloading and maximize resource usage.

Visualizing Statistical Significance: Figure 12 provides a comprehensive overview of the significant results from the ANOVA and Kruskal-Wallis tests. Metrics with p-values below 0.05 are considered statistically significant. The x-axis shows the test method (ANOVA or Kruskal-Wallis), while the y-axis lists the gameplay metrics. This visualization highlights the most critical differences between player clusters, emphasizing which aspects of gameplay and player behavior were most distinct across groups.

The MBTI classification of players did not yield any meaningful insights. The primary reason for this limitation is the small sample size. With only 13 participants and 16 possible personality types, not only were several personality types not represented, but the number of participants per type was also too low to generate statistically significant results.

These findings highlight the strong correlation between strategic decision-making, reaction speed, and overall game success. By leveraging statistical analysis, this study demonstrates that player performance is not solely dictated by raw skill but is also influenced by individual playstyles and decision-making tendencies. The insights gained from this research lay the groundwork for future applications in adaptive game design, where difficulty and game mechanics could be dynamically adjusted based on a player's identified behavioral patterns.

4.2 Time Series Correlation Analysis

This section examines the correlation between player behavior and physiological responses over time, providing insights into how stress levels, heart rate, and gameplay events are interlinked. This analysis is particularly relevant for understanding how players react to in-game challenges and whether their physiological responses align with their gameplay strategies.

To conduct this analysis, the Pearson correlation coefficient was used to measure the relationship between real-time biometric data

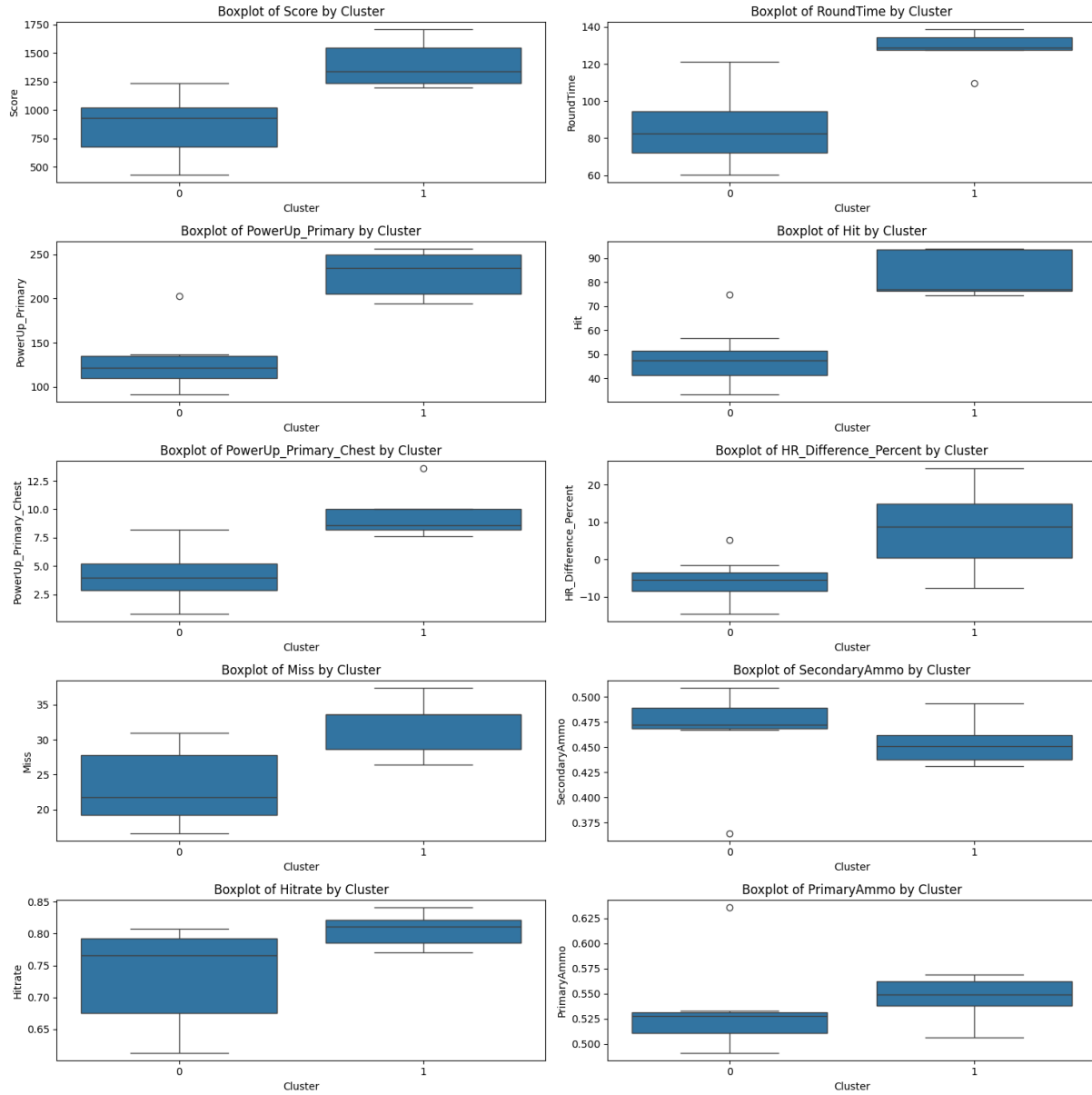


Figure 8: Boxplot comparison of selected features across clusters created through agglomerative performance-based clustering. Differences in point accumulation, round duration, and physiological responses are visible between groups.

and gameplay metrics. The oxygen saturation was constant, so it was not longer considered in further analysis. Each player's physiological data (heart rate recordings) was synchronized with in-game telemetry data to ensure accurate comparisons. Instead of aggregating the data for each player, each gameplay session was analyzed individually, considering chronological sequences of player actions and their corresponding physiological responses. Because heart rate changes may not appear exactly at the same time as the game events, a *lag* was implemented to analyze later reactions. For example a *lag* of two investigates if the heart rate changed two seconds after the gameplay event occurred.

As can be seen in Figure 13. The analysis revealed strong correlations between heart rate fluctuations and intense in-game moments, such as boss spawn or survival-critical situations when the player was in danger of losing a lifepoint. Players who engaged in aggressive playstyles exhibited higher stress levels, with heart rate spikes corresponding to moments of high enemy density or imminent danger.

Another key observation was that players who adopted a defensive playstyle showed less pronounced fluctuations in their physiological responses, suggesting that they maintained a more controlled and measured approach to the game. This aligns with

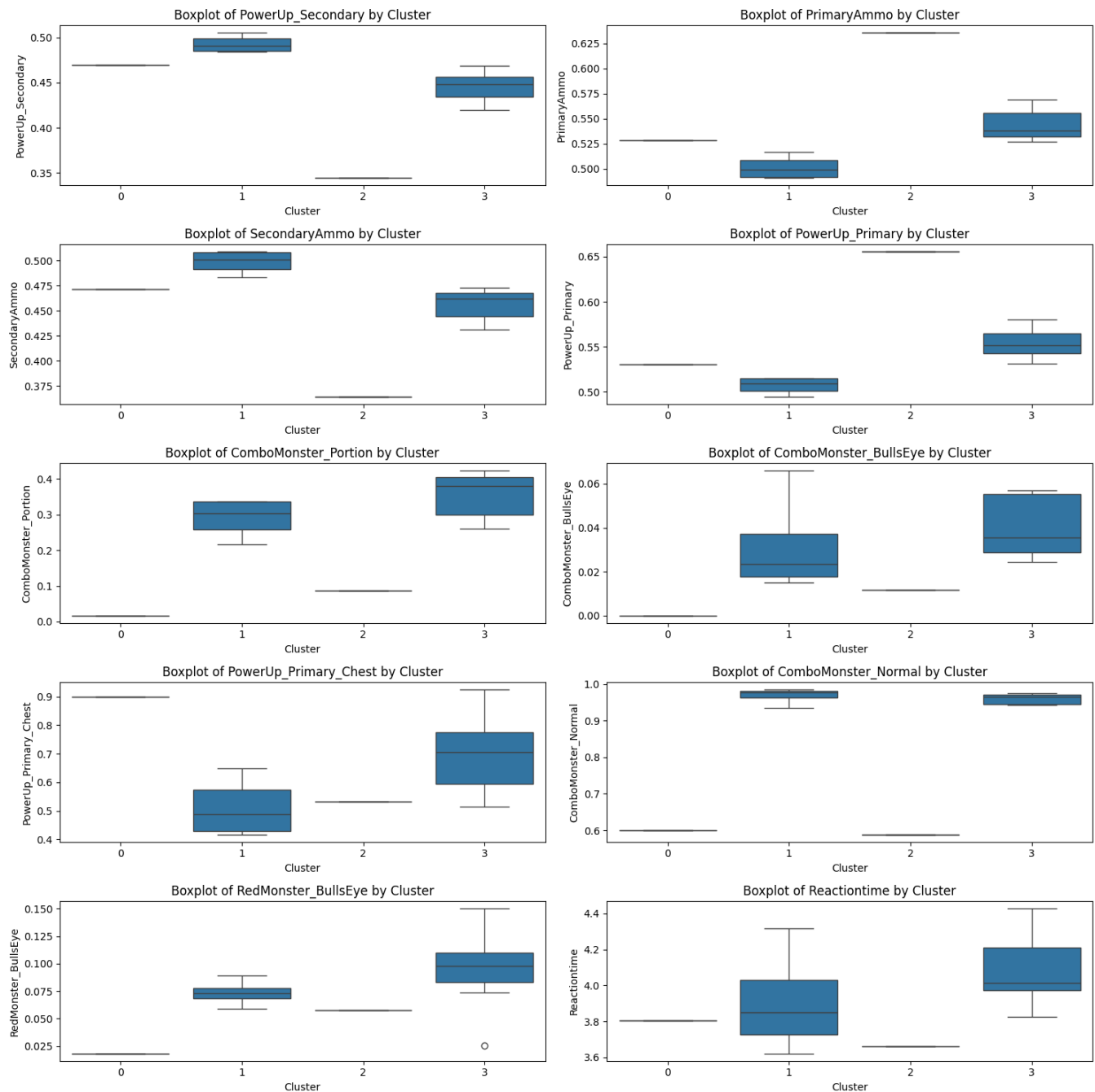


Figure 9: Comparison of the four player clusters based on strategy-focused features. Differences are especially visible in ammo usage and monster prioritization.

findings from the clustering analysis, where defensive players prioritized survival over high-risk, high-reward gameplay strategies.

These results highlight the potential for integrating biometric feedback into adaptive AI systems, enabling games to dynamically adjust difficulty, pacing, or game mechanics based on real-time

player stress levels. Future work could expand on this by developing real-time adaptive mechanisms that respond to biometric data, creating a more immersive and player-centric gaming experience.

Summary of Key Findings

The main findings of this study can be summarized as follows:

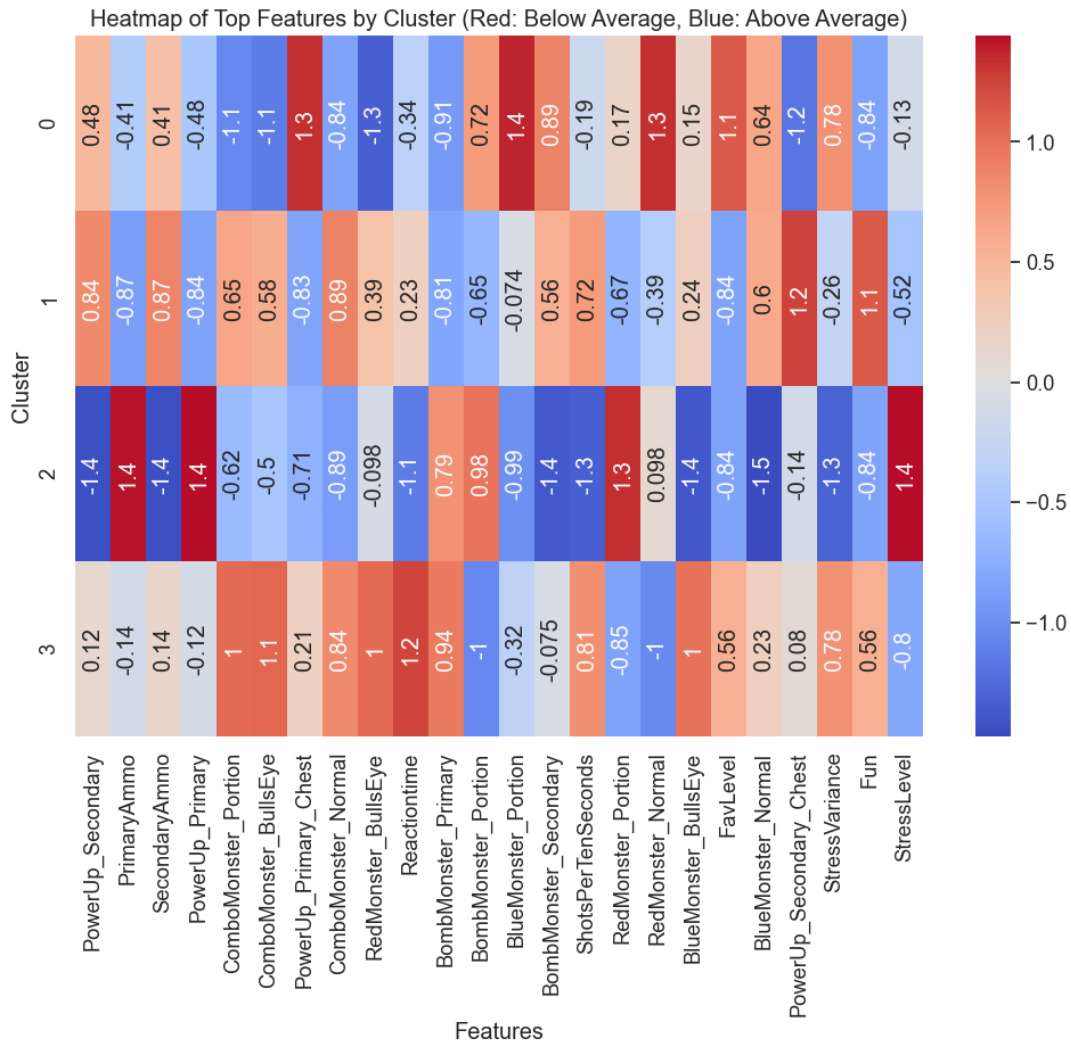


Figure 10: Heatmap showing normalized feature values across strategy-based clusters. Blue means above average, red means below average.

- **Clustering of Player Behavior:** Using K-Means and Agglomerative Clustering, players were successfully grouped based on gameplay telemetry and physiological signals. These clusters revealed distinct playstyles and interaction patterns.
- **Performance and Strategic Differences Between Clusters:** Clusters differed in key performance indicators, including hit accuracy, weapon usage efficiency, and reaction speed. Some groups adapted better to escalating difficulty or managed combo and threat enemies more effectively.
- **Physiological Stress Responses:** Several player groups exhibited distinct physiological patterns, such as elevated heart rate or reduced heart rate variability during stress-inducing scenes (e.g., when managing threat monsters). These trends indicate differentiated stress responses under gameplay pressure.
- **Exploratory Trends in Personality and Playstyle:** Participants completed an MBTI-based personality survey prior to gameplay. While this data was collected to explore potential links between personality type and in-game behavior, the limited and uneven distribution of MBTI types prevented statistically meaningful analysis. As such, no reliable correlations could be established. However, the integration of personality measures into the clustering pipeline remains a promising direction for future studies with larger sample sizes.
- **Potential for Adaptive Game Design:** The results demonstrate that combining biometric and behavioral data enables meaningful player profiling. These profiles can support adaptive game mechanics that dynamically adjust difficulty or feedback to match individual stress tolerance and strategic tendencies.

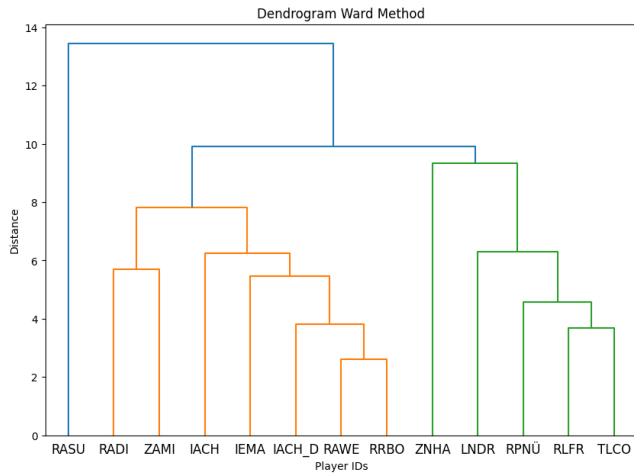


Figure 11: Dendrogram showing strategy-based player clusters. Branch length represents differences in playstyle.

5 Discussion

This study investigated how player behavior can be classified using telemetry data, physiological responses, and subjective feedback to establish a data-driven foundation for adaptive gaming systems. The clustering analysis effectively identified distinct player types, revealing that success in the game is influenced not only by raw skill but also by strategic decision-making and playstyle.

Physiological data provided further insights, showing that heart rate fluctuations correlated with different in-game scenarios and player strategies. These findings suggest that real-time biometric feedback could be a valuable tool in enhancing adaptive game mechanics, enabling difficulty and pacing adjustments based on stress and engagement levels.

The results highlight the potential of machine learning-based clustering techniques in adaptive gaming. Unlike traditional heuristic-based difficulty scaling, which relies on predefined performance thresholds, unsupervised learning methods such as K-Means and Agglomerative Clustering offer a more precise, behavior-driven approach. This study demonstrates that player classification can extend beyond in-game performance metrics by incorporating telemetry data, physiological responses, and subjective feedback to provide a deeper understanding of player engagement.

Furthermore, the correlation between physiological responses and gameplay behavior suggests that biometric feedback could enhance adaptive systems. In the future, real-time monitoring of heart rate and stress levels may allow dynamic adjustments to pacing, enemy difficulty, or task complexity, ensuring that the game remains challenging yet enjoyable for each player.

Despite these contributions, this study has several limitations. The relatively small participant pool affects the generalizability of the findings, and the simplicity of the game prototype limits the scope of gameplay behaviors that can be analyzed. Future research should expand the dataset to include a more diverse player base

and explore a wider range of physiological signals, such as electrodermal activity (EDA) or EEG, to gain deeper insights into player engagement and cognitive load.

Future research directions should focus on implementing and testing real-time AI-driven difficulty adjustments using the player classifications identified in this study. The clustering results indicate that adaptive AI could personalize game mechanics by adjusting difficulty, resource availability, and enemy behavior based on a player's strategic approach. Additionally, deep learning methods could be explored for even more accurate predictions of player behavior, allowing adaptive AI to refine difficulty scaling dynamically.

By integrating these insights, future adaptive AI systems could significantly enhance player engagement, satisfaction, and retention, paving the way for more intelligent and personalized gaming experiences. As real-time biometric monitoring becomes more accessible, AI-driven difficulty adjustments could become a standard feature in modern game development, ensuring that games remain challenging, immersive, and tailored to individual player preferences.

Limitations

This study was designed as an exploratory investigation, and its results should be interpreted within that context. The small sample size ($n = 13$) limits the generalizability and statistical power of the findings, particularly in relation to the personality data. While trends could be observed across clusters, no robust claims can be made regarding the broader player population. Additionally, the use of a simplified, custom-built prototype game helped ensure consistent data collection but may have reduced ecological validity. Real-world games often involve more complex mechanics and unpredictable player behavior, which could influence the relevance of the observed patterns. Future work should replicate this study with larger and more diverse participant groups and test the framework across different game genres to validate its adaptability.

6 Conclusion

This study investigated how player behavior can be classified using telemetry data, physiological responses, and subjective feedback, providing a foundation for future adaptive AI systems in gaming. By employing machine learning techniques, including K-Means and Agglomerative Clustering, distinct player groups were identified based on performance metrics and strategic decision-making. The results demonstrated that success in the game is not only dependent on raw skill but is also influenced by the play style and decision-making tendencies, highlighting the importance of adaptive game mechanics that accommodate diverse player strategies.

The clustering analysis revealed that players could be categorized into high-performing, survival-focused, and low-performing groups based on their scores and survival times. Further segmentation based on strategy indicated clear aggressive and defensive playstyles, with resource management and attack frequency emerging as key differentiators. Unlike prior work that focused mainly on performance or skill-based metrics, this study introduces strategy-based clustering as a novel approach to reveal how players choose to interact with the game, independent of their outcome. This allows for the identification of personal preferences, such as focusing

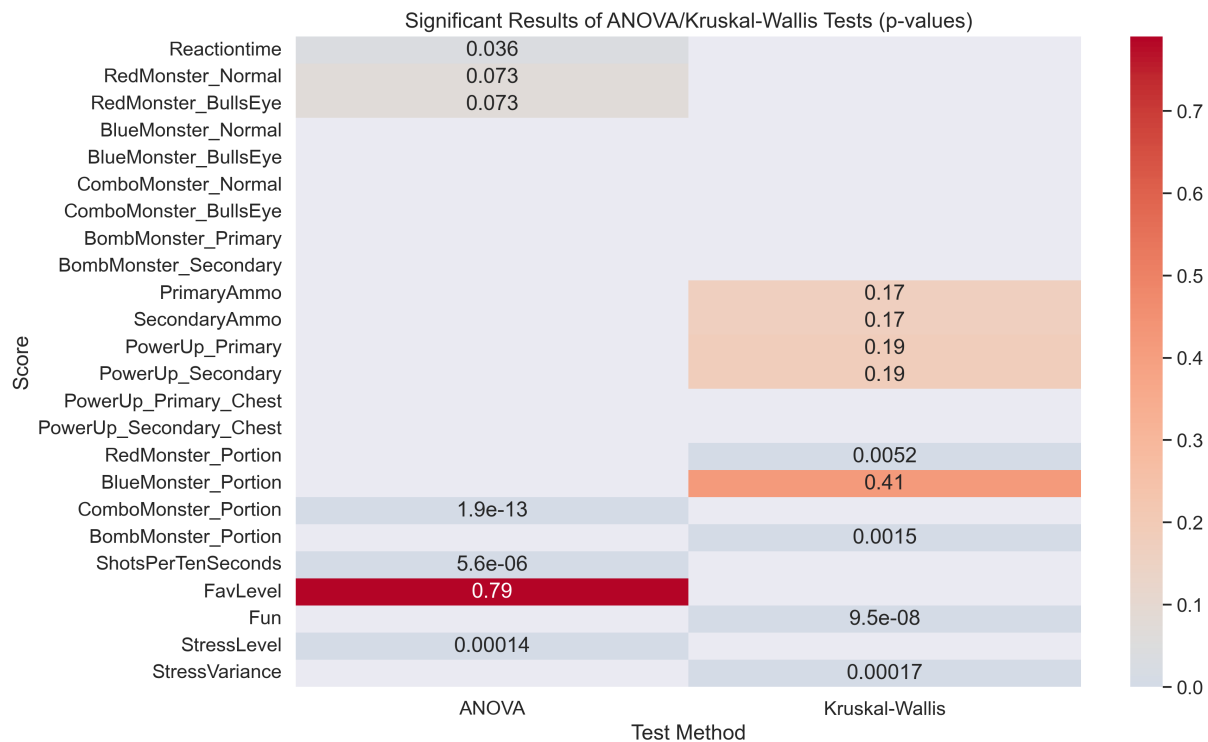


Figure 12: Heatmap displaying the significant results of ANOVA and Kruskal-Wallis tests for various gameplay metrics. The color intensity represents the p-value, with darker colors indicating higher significance. The x-axis shows the test method (ANOVA or Kruskal-Wallis), and the y-axis lists the gameplay metrics. Metrics with p-values below 0.05 are considered statistically significant.

on a single weapon due to cognitive load—an insight that cannot be observed through performance data alone.

Additionally, physiological data analysis showed that heart rate fluctuations corresponded to in-game stress levels, reinforcing the potential use of biometric feedback in dynamic difficulty adjustment. The integration of real-time physiological signals into the clustering process provides an additional dimension for understanding player engagement, offering more personalized adaptation opportunities beyond traditional heuristics.

While the study attempted to correlate MBTI personality types with player behavior, the small sample size prevented meaningful conclusions. However, the inclusion of personality data represents an important step toward holistic player profiling, showing the potential of combining psychological, physiological, and gameplay data in future work.

In conclusion, this study demonstrates that integrating game telemetry, physiological responses, and subjective feedback enhances player classification, offering a more precise foundation for adaptive AI systems. The proposed methodology moves beyond fixed difficulty models by capturing how players act, react, and perceive gameplay situations, making it possible to tailor experiences on a deeper level. While adaptive AI was not implemented, the research provides a structured methodology for future game adaptation systems. By showing that players can be classified not

only by performance but also by behavior and emotional responses, this work lays the groundwork for adaptive AI that is more human-centered, flexible, and context-aware. By advancing real-time player classification and integrating biometric-driven adaptations, game developers can create more engaging, personalized, and responsive gaming experiences that dynamically adjust to player behavior and preferences.

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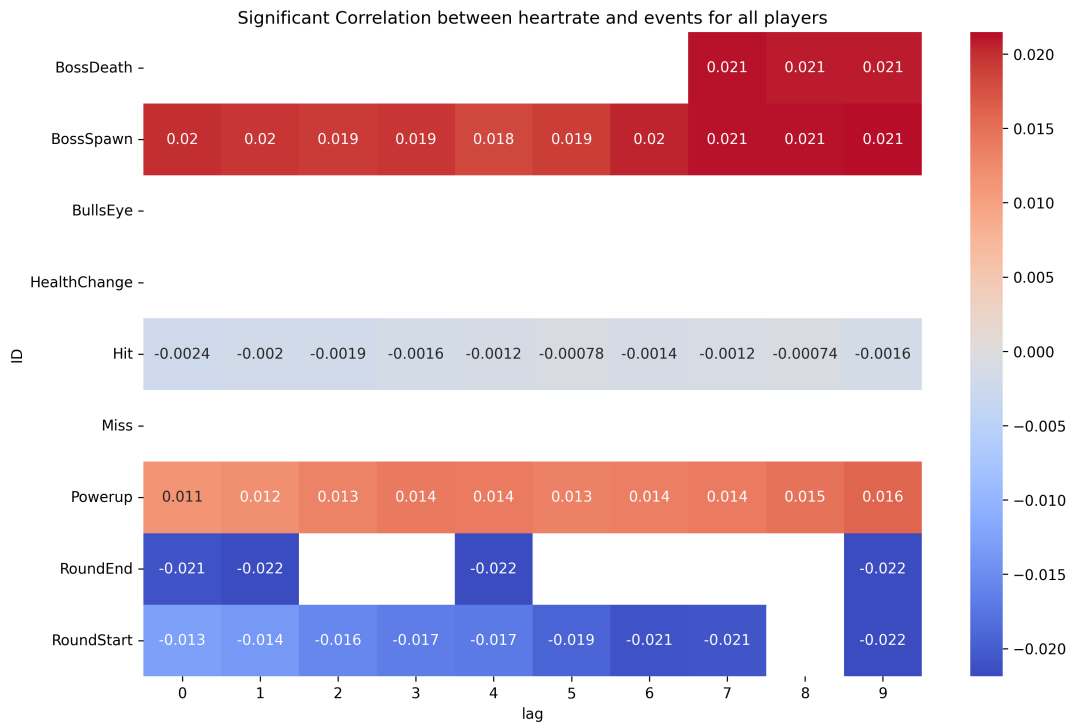


Figure 13: Heatmap showing the significant correlations between heart rate changes and various game events at different time lags. Positive correlations (in red) indicate that heart rate increases after the event, while negative correlations (in blue) indicate a decrease. The x-axis shows the time lag after the event (in seconds or units), and the y-axis shows the type of in-game event. Only statistically significant correlations ($p < 0.05$) are displayed.

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